



University Institute of Engineering
Department of Computer Science & Engineering

Experiment-10

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1. Aim of the practical: To study and implement **PostgreSQL Transactions** using **COMMIT, ROLLBACK, SAVEPOINT, and exception handling** to maintain data integrity and control database operations.

2. Questions

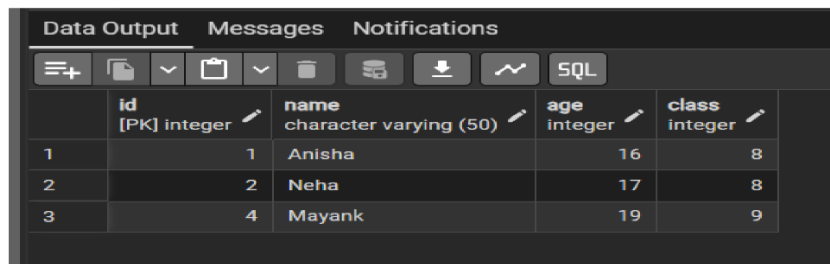
(A) Design a robust PostgreSQL transaction system for the students table where multiple student records are inserted in a single transaction. If any insert fails due to invalid data, only that insert should be rolled back while preserving the previous successful inserts using savepoints. The system should provide clear messages for both successful and failed insertions, ensuring data integrity and controlled error handling.

```
DO $$
BEGIN TRANSACTION
-- Start a transaction
BEGIN
-- Insert multiple students
INSERT INTO students(name, age, class) VALUES ('Anisha',16,8);
INSERT INTO students(name, age, class) VALUES ('Neha',17,8);
INSERT INTO students(name, age, class) VALUES ('Mayank',19,9);

-- If all succeed
RAISE NOTICE ' Transaction Successfully Done';

EXCEPTION WHEN OTHERS THEN
-- If any insert fails
RAISE NOTICE 'Transaction Failed..! Rolling back changes.';
RAISE; -- this will rollback the entire transaction
END;
END;
$$;
```

Solution:



	id [PK] integer	name character varying (50)	age integer	class integer
1	1	Anisha	16	8
2	2	Neha	17	8
3	4	Mayank	19	9

3. Learning Outcomes:

- Learned to create and manage transactions in PostgreSQL.
- Applied COMMIT and ROLLBACK for controlling database changes.
- Learned to use SAVEPOINT for partial rollback within a transaction.
- Understood exception handling and controlled error management.
- Developed practical skills in maintaining data integrity during transactions.